SELF-SUPERVISED LEARNING

Challenges in medical data annotation

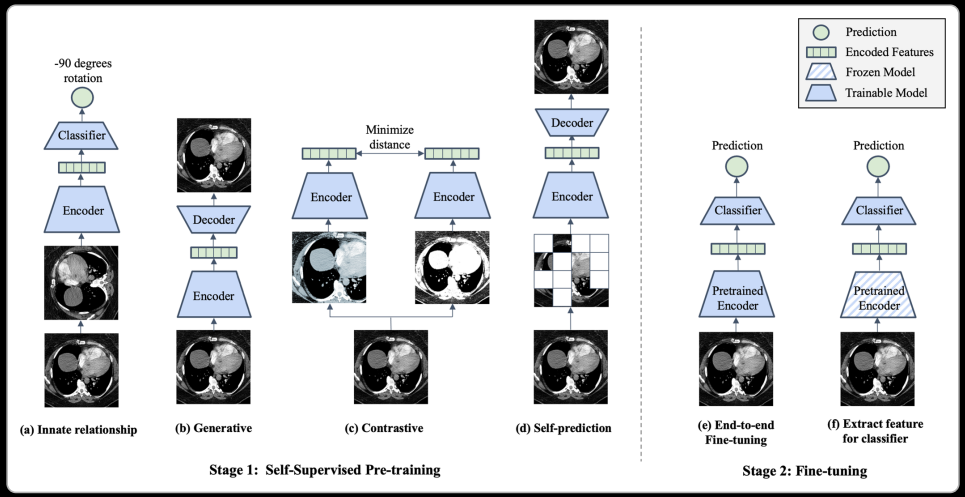
- Traditional manual annotation for complex medical imaging can take radiologists **months**.
- Healthcare organisations were projected to generate **2,314 exabytes** of data in 2020.
- Less than **3%** of this data is properly annotated for AI training.

Medrano, *The $1.5 Trillion AI Healthcare Revolution: Why Data Annotation Is the Silent Engine Driving Medical Superintelligence*, LinkedIn

What is SELF-SUPERVISED learning?

- Supervised learning: train directly on the task, using expert labels.
- Self-supervised learning: learn an encoder from a **pretext task** whose answer comes from the data itself, then fine-tune it with far fewer labels.
- Advantages: self-supervised learning uses large-scale unlabelled data to learn transferable representations across images, text, and speech, reducing annotation costs and potentially improving generalisation when labelled data are limited.

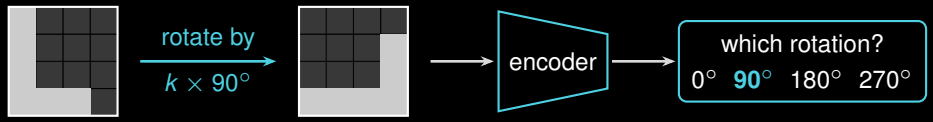
EXAMPLES OF PRETEXT TASKS



Huang et al., *Self-supervised learning for medical image classification: a systematic review and implementation guidelines*, npj Digital Medicine

Innate relationship

PREDICT THE TRANSFORMATION

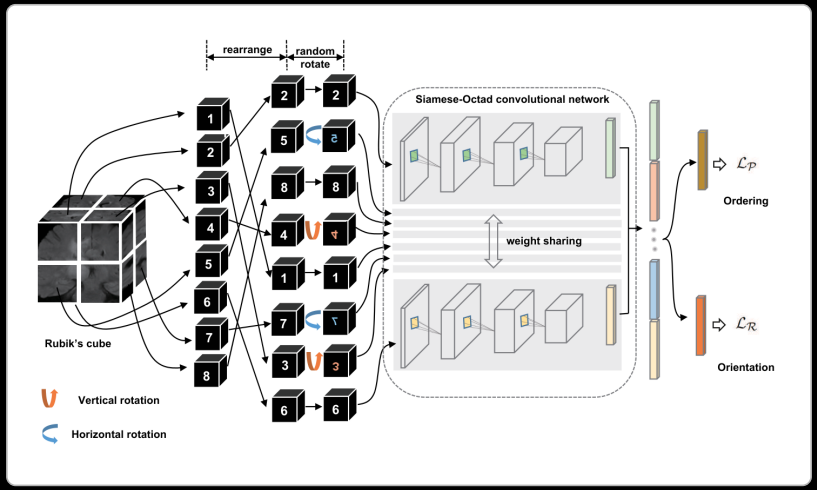


unlabelled scan

- Innate relationship prediction trains a model to maintain its understanding of a data sample after it is transformed in some way.
- For example, rotating an input image and tasking a model with predicting the change degree and direction of rotation relative to the original input.

Huang et al., *Self-supervised learning for medical image classification: a systematic review and implementation guidelines*, npj Digital Medicine

RUBIK'S CUBE RECOVERY

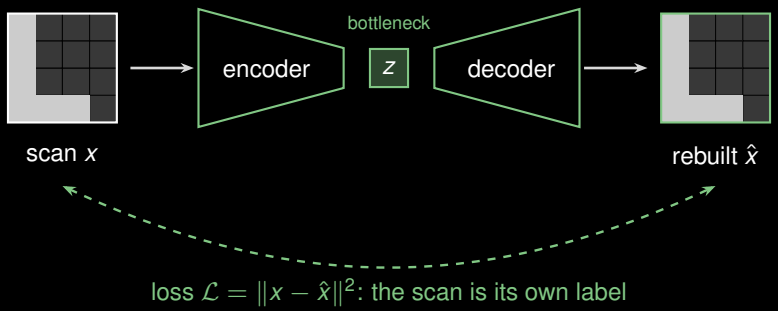


- **Create a 3D puzzle**: Divide a brain scan into eight cubes ($2 \times 2 \times 2$), shuffle their positions, and randomly rotate some of them.
- **Learn without labels**: Train the network to recover the original positions and orientations, encouraging it to learn the spatial structure and anatomy of the brain.
- **Improve downstream performance**: Pretraining on this task outperforms training from scratch on brain hemorrhage classification and brain tumor segmentation.

Zhuang et al., *Self-supervised Feature Learning for 3D Medical Images by Playing a Rubik's Cube*, MICCAI

Generative

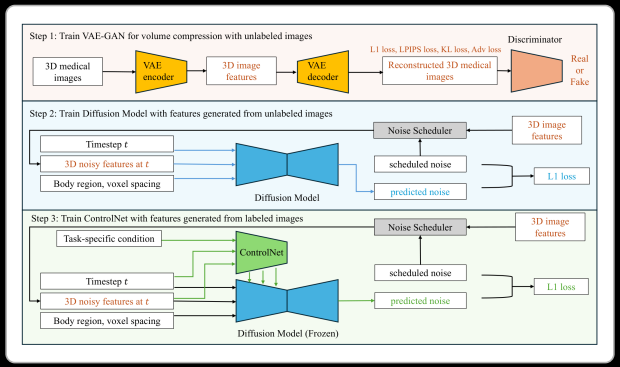
GENERATIVE MODEL



- Generative approaches, including autoencoders, variational autoencoders, and generative adversarial networks (GANs), learn patterns in training data that allow them to reconstruct inputs or generate new, similar samples.
- These methods can learn useful latent representations without manually annotated labels, drawing their training signals from the data itself.
- Reconstruction-based methods offer a clear example of self-supervised learning: the original input serves as the target, teaching the model to capture the information needed to recreate it.

Huang et al., *Self-supervised learning for medical image classification: a systematic review and implementation guidelines*, npj Digital Medicine

What MAISI learns first

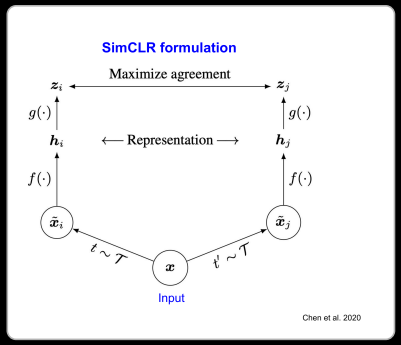


- **Step 1**: the volume compression network (VAE-GAN) is trained on 39,206 3D CT volumes and 18,827 3D MRI volumes. It compresses high-resolution 3D medical images into a latent space that is perceptually equivalent to the image space.
- **Step 2**: a latent diffusion model is trained on 10,277 CT volumes. It operates within the compressed latent space, conditioned on specific body regions and voxel spacing, to generate features of realistic and complex 3D anatomical structures.
- **Step 3**: ControlNet is integrated into the MAISI framework. It allows dynamic control over the generated outputs by injecting additional conditions into the trained latent diffusion model.
- In Module 2 we used MAISI to make data; this is what it had to learn first.

Guo et al., *MAISI: Medical AI for Synthetic Imaging*, WACV

Contrastive

SimCLR and its loss



$$\ell_{i,j} = -\log \frac{\exp\big(\mathsf{sim}(z_i, z_j)/\tau\big)}{\sum_{k \neq i} \exp\big(\mathsf{sim}(z_i, z_k)/\tau\big)}$$

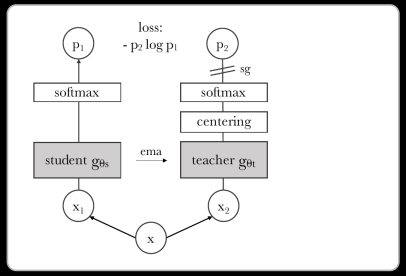
sim: cosine similarity; τ: temperature

SimCLR has four components:

- **Data augmentation**: transform one image randomly in two ways, giving two views of the same example.
- **Base encoder** (e.g. ResNet-50): turns each view into a representation vector. This is the network you keep for the downstream task.
- **Projection head**: a small MLP that maps the representation to a smaller vector for the loss.
- **Contrastive loss**: low when the two views of the same image are similar and the other images in the batch (the *negatives*) are dissimilar; high otherwise.

Chen et al., *A Simple Framework for Contrastive Learning of Visual Representations*, ICML

DINO, DISTILLATION WITH NO TEACHER TO DISTIL FROM

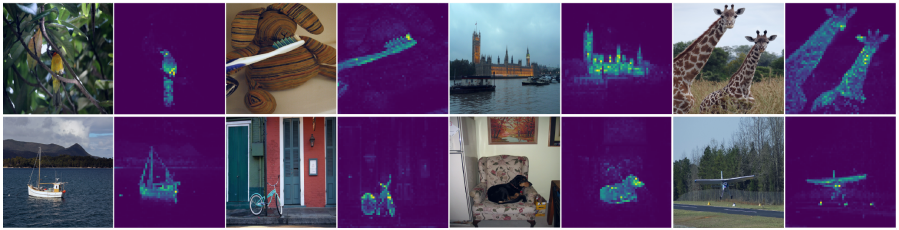


Two views of x; the student is trained, the teacher is an exponential moving average of it.

- Student and teacher each see a view of the same image; the teacher is an **exponential moving average** of the student.
- Output: a **softmax over K output dimensions**. Loss: **cross-entropy** $H(p_t, p_s) = -\sum p_t \log p_s$, as in knowledge distillation.
- **No negatives**: **centering** stops one dimension dominating; **sharpening** stops a uniform answer.
- Third time the same loss: SimCLR, the Module 1 classifier, and DINO all minimise a cross-entropy.
- With **no labels**: ViT-S/8 reaches **78.3%** on ImageNet (k-NN); ViT-B/8 **80.1%** (linear probe).

Caron et al., *Emerging Properties in Self-Supervised Vision Transformers*, ICCV

DINO ATTENTION MAPS



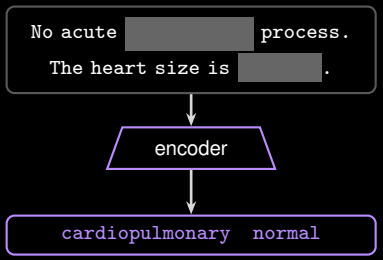
- With **no mask, no box and no label**, DINO's self-attention learns where the object is in each image.

Caron et al., *Emerging Properties in Self-Supervised Vision Transformers*, ICCV

Self-prediction

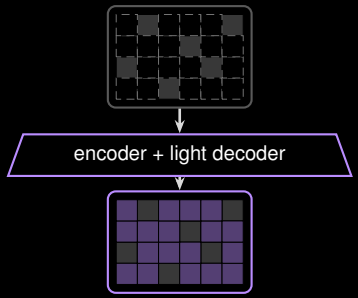
YOU ALREADY USE ONE EVERY DAY

BERT: hide a word, predict it



- **Masked language modeling**: replace a random sample of words with `[MASK]` and predict them, from unannotated text alone.
- BERT is **bidirectional**: it reads the words both before and after the gap.
- Then fine-tune for tasks such as text classification.

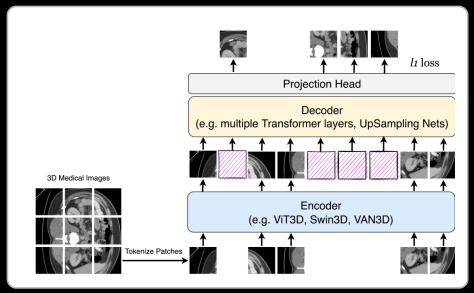
MAE: hide patches, predict them



- **Masked image modeling**: split the image into patches and mask a large subset, commonly **75%**.
- Only the **visible** patches enter the encoder; a lightweight decoder reconstructs the masked content.
- Loss on the masked patches M only: $\mathcal{L} = \frac{1}{|M|} \sum_{i \in M} \|m_i - x_i\|^2$

Murel and Kavlakoglu, *What are masked language models?*, IBM Think; *Masked Image Modeling (MIM)*, Emergent Mind; Devlin et al., *BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding*, NAACL; He et al., *Masked Autoencoders Are Scalable Vision Learners*, CVPR

MASKED IMAGE MODELLING

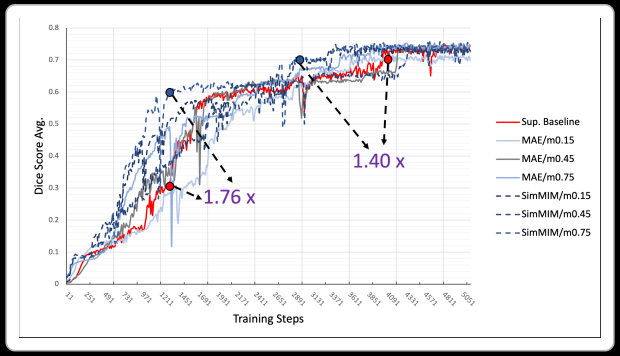


MAE on a 3D CT volume: only the visible patches enter the encoder; a light decoder fills in the hatched ones.

- Divide the image into non-overlapping patches (e.g. a 96^3 CT volume into 216 patches of 16^3).
- Mask patches at random with a **high masking ratio**, so the task cannot be solved by copying visible neighbours.
- Masked patches are invisible to the encoder; only the decoder sees all tokens.
- The loss is computed **only on masked patches**.
- Only the encoder is kept for fine-tuning. SimMIM replaces the decoder with a single projection layer.

Chen et al., *Masked Image Modeling Advances 3D Medical Image Analysis*, WACV; He et al., *Masked Autoencoders Are Scalable Vision Learners*, CVPR; Xie et al., *SimMIM: A Simple Framework for Masked Image Modeling*, CVPR

Masked pre-training on CT



- Masked image modeling approaches outperform contrastive learning methods in general, as both MAE and SimMIM achieve an average dice score of around 0.752$\sim$0.758, while SimCLR achieves an average dice score of around 0.723, which is 4.5% lower than the best approach.
- Masked image modeling approaches dramatically increase the training speed and reduce the cost.
- SimMIM based architectures can obtain a 1.76$\times$ better dice score at the 1.3k training step.
- MIM based approaches can reach a dice score of 0.7 with 1.4$\times$ less training time than the training time required for supervised baseline.

Chen et al., *Masked Image Modeling Advances 3D Medical Image Analysis*, WACV